

Table S1. Legend information for the basal metabolic rate (BMR) database provided in Table S2.

	Column	Column name	Description
Information concerning the species studied	A	No	Entry number
	B	Species W&R	Species name following Wilson & Reeder (2005).
	C	Subsp W&R	Subspecies name when indicated in original work and recognized by Wilson & Reeder (2005).
	D	Species original	Species name in original work.
	E	Species Nexus	Species name in phylogenetic tree (see File S4).
	F	Order	Order name abbreviated.
	G	Domestic	1 = domestic; 0 = wild (NA = 'not available', unknown whether domestic or not).
Source	H	Authors	Original authors of the metabolic value.
	I	Cited value	1 = metabolic value reported in the literature as 'unpublished data'. In most of these cases, no experimental information is given, thus a complete evaluation is impossible. The authors who reported the value are listed in column K (secondary reference). 0 = value obtained by the authors cited in column H.
	J	Source unavailable	1 = the original source was unavailable and could not be examined, usually preventing an evaluation of the metabolic value. 0 = the original source was examined.
	K	Secondary reference	When the metabolic value was reported in the literature as 'unpublished data' (cited value = 1) or when the original publication was unavailable (source unavailable = 1), the authors who cited the metabolic value are given.
Body mass and BMR data	L	mass	Body mass in g (NA = 'not available').
	M	BMR	Estimate of basal rate of metabolism in ml O ₂ /h (NA = 'not available').
	N	BMR %	BMR as a percentage of the value expected from the equation $BMR = 2.382 m^{0.728}$ (allometric equation obtained with PGLS for the SELECT values on mammals) (NA = 'not available').
	O	a	Sample size (NA = 'not available').
Standard BMR criteria	P	b	1 = a graph of rate of metabolism as a function of ambient temperature is provided; 0 = a graph is not provided.
	Q	b1	1 = a graph of rate of metabolism as a function of ambient temperature is available in the literature for this particular species; 0 = no graph is available for this species.
	R	c	1 = measurements were done at thermal neutrality; 0 = some doubt exists about whether the measurements were done at thermal neutrality; -1 = measurements were not done at thermal neutrality.
	S	d	1 = animals were measured at rest; 0 = some doubt exists about whether the animals were resting; -1 = animals were not resting.
	T	e	1 = measurements were done on postabsorptive animals; 0 = some doubt exists about whether the animals were postabsorptive; -1 = animals were not postabsorptive.
	U	f	1 = measurements were done during a major resting phase of the daily cycle; 0 = some doubt exists about whether the measurements were done during a major resting phase of the daily cycle; -1 = measurements were not done during a major resting phase of the daily cycle.
	V	g	1 = adult animals; 0 = some doubt exists about whether the animals were adults; -1 = animals were not adult.
	W	h	1 = females were neither pregnant nor lactating; 0 = some doubt exists about whether the females were non-reproductive; -1 = at least some females were pregnant or lactating.
Additional concerns	X	i	1 = additional concerns about the species studied: the animals measured are difficult to assign safely to a particular taxon.
	Y	j	1 = additional concerns about animal origin: the animals were held for a long time under climatic conditions which could have affected BMR (e.g. some zoo or pet animals, or individuals originating from long-term captive colonies held in artificial conditions). The animals were submitted to long-lasting experimental acclimation (temperature, photoperiod, diet). The animals originated from lineages submitted to several generations of artificial selection, or from experimentally modified ecosystems.
	Z	k	1 = additional concerns about the physical state of the animals: the animals were in bad condition, nutritionally stressed, or obese.
	AA	l	1 = unexpectedly small body mass; the animals could be juveniles.
	AB	m	1 = additional concerns about animal handling and conditions. The animals were measured in conditions which may be inappropriate for BMR measurements: possibly entering torpor, standing rather than lying, measured within a nest. The animals were submitted to particularly stressful conditions (e.g. very small chamber, probes, bounds, harness). The animals were anaesthetized or sedated or anaesthetized less than 24 h before measurements. Animals with organs removed. Several individuals were measured together.
	AC	n	1 = additional concerns about the set-up and protocol. The respirometer may be inappropriate (e.g. closed system). The entire holding cage was put into the respirometer. Conditions unclear (e.g. experimental temperature not indicated) or set-up unclear (including when described only in an abstract). The experimental protocol may be inappropriate for BMR estimation (very short runs, very short continuous VO ₂ measurements, no continuous VO ₂ measurements or VO ₂ averaged over long measurement periods).
	AD	o	1 = additional concerns about the analysis. Analysis questionable (e.g. thermoneutral zone) and/or calculation unclear or inappropriate. Thermoneutral zone poorly defined. Value imprecise (e.g. extracted from graphs), corrected, recalculated or unclear due to inconsistency (e.g. graphs and numerical values different).
	AE	p	1 = additional concerns about body mass. Body mass is not provided, unclear, recalculated from metabolic values, or does not correspond precisely to the metabolic value. The lack of body mass data precludes recalculation of absolute VO ₂ when the latter is not reported explicitly.
	AF	q	1 = additional concerns about the BMR value reported. Value unexpected compared to other values on the same or on closely related species. Value based on very few measurements, or obtained by pooling data from several studies or from several species or from wild and domestic animals.
Overall evaluation	AG	E_{crit}	Overall four-level evaluation of how the BMR criteria (c–h) were met. E_{crit} is based on C_1 , the number of criteria clearly not being met (–1 entries) and on C_2 , the number of criteria for which there were doubts (0 entries). The following scale was applied for E_{crit} : $E_{crit} = 1$ if $C_1 = C_2 = 0$ $E_{crit} = 2$ if $C_1 = 0$ and $1 \leq C_2 \leq 4$ $E_{crit} = 3$ if $C_1 = 0$ and $C_2 \geq 5$, or if $C_1 = 1$ and $C_2 \leq 1$ $E_{crit} = 4$ for all other cases
	AH	E	Overall four-level evaluation of the data as a BMR estimate (NA = 'not available', when no evaluation was possible). Initially, E was set equal to E_{crit} . In a second stage, it was corrected by taking into account additional problems encountered (l–q). E was elevated by 2 (provided that $E_{crit} \leq 2$) when the severity of the problems identified were sufficient to lead to a rejection of the estimate. It was elevated by 1 (provided that $E_{crit} \leq 3$) when the additional problems were not considered serious enough to justify by them alone a rejection of the estimate. No correction was applied to E when no additional problem was identified or when the additional problems were considered minor enough.
	AI	Accepted	1 = value accepted as a reliable BMR estimate ($E = 1$ or 2); 0 = rejected ($E = 3$ or 4).
Inclusion in specific data sets	AJ	ALL	1 = value included in the ALL data set: value considered as the best estimate of BMR for this particular species.
	AK	SELECT	1 = value included in the SELECT data set: best accepted estimate of BMR for this particular species.
	AL	SPOILED	1 = value included in the SPOILED dataset: either an accepted BMR estimate (species with a single accepted estimate) or a worst estimate, most divergent from the best accepted estimate (for species represented by several entries, of which at least one is an accepted value).